

API SPECIFICATION & ENDPOINTS

For: 400k Users in Cameroon | Concurrent: 1,000 | Date: 8/17/2026 | Status: PRODUCTION-READY

API SPECIFICATION (COMPLETE)

Base URL

https://videocall.example.com/api

WebSocket: wss://videocall.example.com/ws

Authentication Header

Authorization: Bearer <JWT_ACCESS_TOKEN>

Auth Endpoints

POST /auth/login

Request:

```
{
  "email": "user@example.com",
  "password": "password123"
}
```

Response (200):

```
{
  "access_token": "eyJhbGc...",
  "refresh_token": "refresh...",
  "expires_in": 900,
  "user": {
    "id": "uuid",
    "email": "user@example.com",
    "first_name": "John",
    "last_name": "Doe"
  }
}
```

POST /auth/refresh

Request:

```
{ "refresh_token": "refresh..." }
```

Response (200):

```
{
  "access_token": "eyJhbGc...",
  "expires_in": 900
}
```

```
}
```

POST /auth/logout

Response (204): No content

Call Endpoints

POST /calls/create

Request:

```
{
  "initiator_id": "uuid",
  "call_type": "p2p" // or "group"
}
```

Response (201):

```
{
  "call_id": "uuid",
  "room_id": "CALL_ABC123",
  "sfu_instance": "sfu-1",
  "ice_servers": [
    {
      "urls": ["stun:turn.cameroon.local:3478"],
      "username": "videocall",
      "credential": "password"
    }
  ]
}
```

GET /calls?limit=20&offset=0&user_id=<uuid>

Response (200):

```
{
  "calls": [
    {
      "id": "uuid",
      "initiator_id": "uuid",
      "call_type": "p2p",
      "room_id": "CALL_ABC123",
      "start_time": "2024-01-15T10:30:00Z",
      "end_time": "2024-01-15T10:45:30Z",
      "duration_seconds": 930,
      "recording_id": "uuid",
      "participants_count": 2
    }
  ],
  "pagination": { "total": 100, "limit": 20, "offset": 0 }
}
```

POST /calls/:id/end

Request: {}

Response (200):

```
{
  "call_id": "uuid",
  "call_state": "ended",
  "duration_seconds": 930
}
```

Recording Endpoints

GET /recordings?limit=20&call_id=<uuid>

Response (200):

```
{
  "recordings": [
    {
      "id": "uuid",
      "call_id": "uuid",
      "file_path": "/recordings/call_abc.mp4",
      "file_size_bytes": 52428800,
      "duration_seconds": 930,
      "codec": "h264",
      "bitrate_kbps": 3000,
      "created_at": "2024-01-15T10:45:30Z",
      "expires_at": "2024-02-14T10:45:30Z",
      "status": "complete"
    }
  ]
}
```

GET /recordings/:id/download

Response: Binary MP4 stream (Content-Type: video/mp4)

DELETE /recordings/:id

Response (204): Deleted
